Operating Systems Fundamentals

1. Processor Types

5 Marks Answer

A processor (CPU) is the brain of the computer — it executes instructions and performs calculations.

Types of Processors:

Single-core Processor

- Has only one core (CPU unit).
- Can perform one task at a time.
- Example: Early Intel Pentium.

Multi-core Processor

- Has two or more cores.
- Can run multiple tasks in parallel.
- Example: Intel i5, i7.

Distributed Processor

- Uses multiple processors connected by a network.
- Tasks are divided among processors.
- Example: Cloud servers, clusters.

10 Marks Answer

A processor executes instructions of programs.

Depending on how many processing units (cores) it has and how it handles data, processors are divided as:

Single-core Processor

- Performs one operation at a time. Slower for multitasking.

Multi-core Processor

- Multiple cores work in parallel — better multitasking and speed.

Parallel Processor

- Tasks are broken into smaller sub-tasks and run together.

Distributed Processor Systems

- Uses multiple computers linked through networks. Example: Supercomputers.

Special Purpose Processors

- Designed for specific tasks like GPU (graphics) or DSP (signal).

2. CPU Scheduling and Algorithms

5 Marks Answer

CPU Scheduling is the process of deciding which process will use the CPU next when there are multiple ready processes.

Goals of Scheduling:

- Maximize CPU use
- Minimize waiting and response time
- Improve throughput (number of processes completed)

Common Scheduling Algorithms:

FCFS (First Come First Serve)

- Simple, non-preemptive.
- Process that arrives first is executed first.

SJF (Shortest Job First)

- Process with the smallest burst time runs first.

RR (Round Robin)

- Each process gets equal CPU time (time quantum).

Priority Scheduling

- Higher priority process gets CPU first.

10 Marks Answer

CPU Scheduling determines the order in which processes execute.

When several processes are in the ready queue, the OS chooses one for execution.

Objectives:

- Efficient CPU utilization
- Fairness to all processes
- Minimize turnaround, waiting, and response times

Major Algorithms:

FCFS (First Come First Serve)

- Non-preemptive.
- Simple but causes convoy effect (slow process delays others).

SJF (Shortest Job First)

- Selects the shortest next process.
- Gives best average waiting time.
- Hard to know process length in advance.

Priority Scheduling

- CPU given to highest priority process.
- Can cause starvation (low priority may wait forever).

Round Robin (RR)

- Each process gets a fixed time quantum.
- After that, it moves to the end of the queue.
- Good for time-sharing systems.

Multilevel Queue Scheduling

- Processes grouped by type (system, user, background).
- Each queue has its own scheduling policy.

3. Structure of Operating System

5 Marks Answer

Structure of OS means how its components are organized.

Main Types:

- **Simple Structure** – No proper separation (e.g., MS-DOS).
- **Layered Structure** – OS divided into layers.
- **Monolithic Structure** – All functions in one kernel.
- **Microkernel** – Small kernel; user services run in user mode.
- **Modular Structure** – Combination of monolithic and microkernel.

10 Marks Answer

The structure of an operating system defines how components (like memory, process, I/O, file management) interact.

Types:

Monolithic System:

- Entire OS in one large program (e.g., UNIX).
- Fast but hard to maintain.

Layered System:

- Divided into layers (hardware → user).
- Each layer depends only on the layer below it.

- Easier to debug.

Microkernel System:

- Only basic features (communication, process control) in kernel.
- Other services in user mode.
- Example: macOS, QNX.

Modular System:

- Kernel built using modules that can be added or removed dynamically.
- Example: Linux.

4. Preemptive and Non-Preemptive Scheduling

5 Marks Answer

Preemptive Scheduling:

- CPU can take control from a process if another higher-priority process arrives.
- Example: Round Robin, Priority Scheduling.

Non-Preemptive Scheduling:

- Once a process gets CPU, it cannot be interrupted until it finishes.
- Example: FCFS, SJF (non-preemptive).

10 Marks Answer

Difference Between Preemptive and Non-Preemptive Scheduling

Feature	Preemptive	Non-Preemptive
Control	CPU can be taken away	CPU runs till process ends
Response Time	Better	Poor
Complexity	High	Simple
Examples	RR, Priority	FCFS, SJF
Use Case	Time-sharing systems	Batch systems

Example:
If Process A is running and Process B (high priority) arrives —

- In preemptive, A is paused and B runs.
- In non-preemptive, A continues till it completes.

5. Evolution of Operating System

5 Marks Answer

Evolution of OS means how operating systems improved over time.

Stages:

- **Manual Operating System** – No OS, programs run manually.
- **Batch Processing** – Jobs collected and processed in batches.
- **Multiprogramming** – Many programs loaded; CPU switches between them.
- **Time Sharing** – Multiple users share system interactively.
- **Distributed & Network OS** – Multiple systems connected over network.
- **Modern OS** – GUI, multitasking, mobile & cloud systems.

10 Marks Answer

The Operating System evolved to make computers faster, more efficient, and user-friendly.

Phases of Evolution:

Early Systems (1940s–50s):

- No OS; programs entered manually.

Batch Systems (1950s–60s):

- Jobs collected and run automatically in batches.
- No user interaction.

Multiprogramming (1960s):

- Several programs in memory at once.
- CPU switches to another when one waits for I/O.

Time-Sharing Systems (1970s):

- Many users access the same computer.
- Faster response, interactive.

Distributed Systems (1980s):

- Many computers connected.
- Resources shared over network.

Modern Systems (1990s–Now):

- Support multitasking, security, and GUI.
- Example: Windows, Linux, Android.